

```
# Install a library, for example, pandas
!pip install pandas
```

```
# After installation, you can import and use the library
import pandas as pd
print(f'pandas version: {pd.__version__}')
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
pandas version: 2.2.2
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Data Loading

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("gokulrajkmv/unemployment-in-india")

print("Path to dataset files:", path)
```

Using Colab cache for faster access to the 'unemployment-in-india' dataset.
Path to dataset files: /kaggle/input/unemployment-in-india

```
import os

# The 'path' variable from the previous cell contains the directory where the dataset was downloaded.
# We need to specify the full path to the CSV file.
# A common file name for this dataset is 'Unemployment in India.csv'.
file_path = os.path.join(path, 'Unemployment in India.csv')

# Load the CSV file into a pandas DataFrame named 'data'
data = pd.read_csv(file_path)

# Now, you can display the head of the DataFrame
data.head()
```

1 to 5 of 5 entries [Filter](#) [📄](#) [?](#)

index	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Area
0	Andhra Pradesh	31-05-2019	Monthly	3.65	11999139.0	43.24	Rural
1	Andhra Pradesh	30-06-2019	Monthly	3.05	11755881.0	42.05	Rural
2	Andhra Pradesh	31-07-2019	Monthly	3.75	12086707.0	43.5	Rural
3	Andhra Pradesh	31-08-2019	Monthly	3.32	12285693.0	43.97	Rural

Next steps: [Generate code with data](#) [New interactive sheet](#)

```
data.tail()
```

	Region	Date	Frequency	Unemployment_Rate	Employed	Labour_Participation_Rate	Area	📄
749	West Bengal	2020-02-29	Monthly	7.55	10871168.0	44.09	Urban	
750	West Bengal	2020-03-31	Monthly	6.67	10806105.0	43.34	Urban	
751	West Bengal	2020-04-30	Monthly	15.63	9299466.0	41.20	Urban	
752	West Bengal	2020-05-31	Monthly	15.22	9240903.0	40.67	Urban	
753	West Bengal	2020-06-30	Monthly	9.86	9088931.0	37.57	Urban	

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 7 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Region                                740 non-null    object
 1   Date                                  740 non-null    object
 2   Frequency                             740 non-null    object
 3   Estimated Unemployment Rate (%)       740 non-null    float64
 4   Estimated Employed                    740 non-null    float64
 5   Estimated Labour Participation Rate (%) 740 non-null    float64
 6   Area                                  740 non-null    object
dtypes: float64(3), object(4)
memory usage: 42.1+ KB
```

```
data.describe()
```

	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	
count	740.000000	7.400000e+02	740.000000	
mean	11.787946	7.204460e+06	42.630122	
std	10.721298	8.087988e+06	8.111094	
min	0.000000	4.942000e+04	13.330000	
25%	4.657500	1.190404e+06	38.062500	
50%	8.350000	4.744178e+06	41.160000	
75%	15.887500	1.127549e+07	45.505000	
max	76.740000	4.577751e+07	72.570000	

Data Cleaning

```
data.columns
```

```
Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
      'Estimated Employed', 'Estimated Labour Participation Rate (%)',
      'Area'],
      dtype='object')
```

```
data.columns = data.columns.str.strip()
```

```
data.rename(columns={
    'Estimated Unemployment Rate (%)': 'Unemployment_Rate',
    'Estimated Employed': 'Employed',
    'Estimated Labour Participation Rate (%)': 'Labour_Participation_Rate'
}, inplace=True)
```

```
data['Date'] = pd.to_datetime(data['Date'])
```

```
print(data.isnull().sum())
```

```
Region      0
Date        0
Frequency    0
Unemployment_Rate  0
Employed     0
Labour_Participation_Rate  0
Area        0
dtype: int64
```

```
data.head()
```

	Region	Date	Frequency	Unemployment_Rate	Employed	Labour_Participation_Rate	Area	
0	Andhra Pradesh	2019-05-31	Monthly	3.65	11999139.0		43.24	Rural
1	Andhra Pradesh	2019-06-30	Monthly	3.05	11755881.0		42.05	Rural
2	Andhra Pradesh	2019-07-31	Monthly	3.75	12086707.0		43.50	Rural
3	Andhra Pradesh	2019-08-31	Monthly	3.32	12285693.0		43.97	Rural
4	Andhra Pradesh	2019-09-30	Monthly	5.17	12256762.0		44.68	Rural

Next steps: [Generate code with data](#) [New interactive sheet](#)

```
data.tail()
```

	Region	Date	Frequency	Unemployment_Rate	Employed	Labour_Participation_Rate	Area	
749	West Bengal	2020-02-29	Monthly	7.55	10871168.0		44.09	Urban
750	West Bengal	2020-03-31	Monthly	6.67	10806105.0		43.34	Urban
751	West Bengal	2020-04-30	Monthly	15.63	9299466.0		41.20	Urban
752	West Bengal	2020-05-31	Monthly	15.22	9240903.0		40.67	Urban
753	West Bengal	2020-06-30	Monthly	9.86	9088931.0		37.57	Urban

Exploratory Data Analysis (EDA)

```
print(data.describe())
```

```

count      Date      Unemployment_Rate      Employed  \
count      740      740.000000      7.400000e+02
mean    2019-12-12  18:36:58.378378496      11.787946  7.204460e+06
min      2019-05-31  00:00:00      0.000000  4.942000e+04
25%      2019-08-31  00:00:00      4.657500  1.190404e+06
50%      2019-11-30  00:00:00      8.350000  4.744178e+06
75%      2020-03-31  00:00:00     15.887500  1.127549e+07
max      2020-06-30  00:00:00     76.740000  4.577751e+07
std      NaN      10.721298  8.087988e+06

      Labour_Participation_Rate
count      740.000000
mean      42.630122
min      13.330000
25%      38.062500
50%      41.160000
75%      45.505000
max      72.570000
std      8.111094

```

```
avg_unemployment = data['Unemployment_Rate'].mean()
print("Average Unemployment Rate:", avg_unemployment)
```

```
Average Unemployment Rate: 11.787945945945946
```

```
highest_region = data.groupby('Region')['Unemployment_Rate'].mean().sort_values(ascending=False)
print(highest_region.head())
```

```

Region
Tripura      28.350357
Haryana      26.283214
Jharkhand    20.585000
Bihar        18.918214
Himachal Pradesh  18.540357
Name: Unemployment_Rate, dtype: float64

```

```

plt.figure(figsize=(12,6))

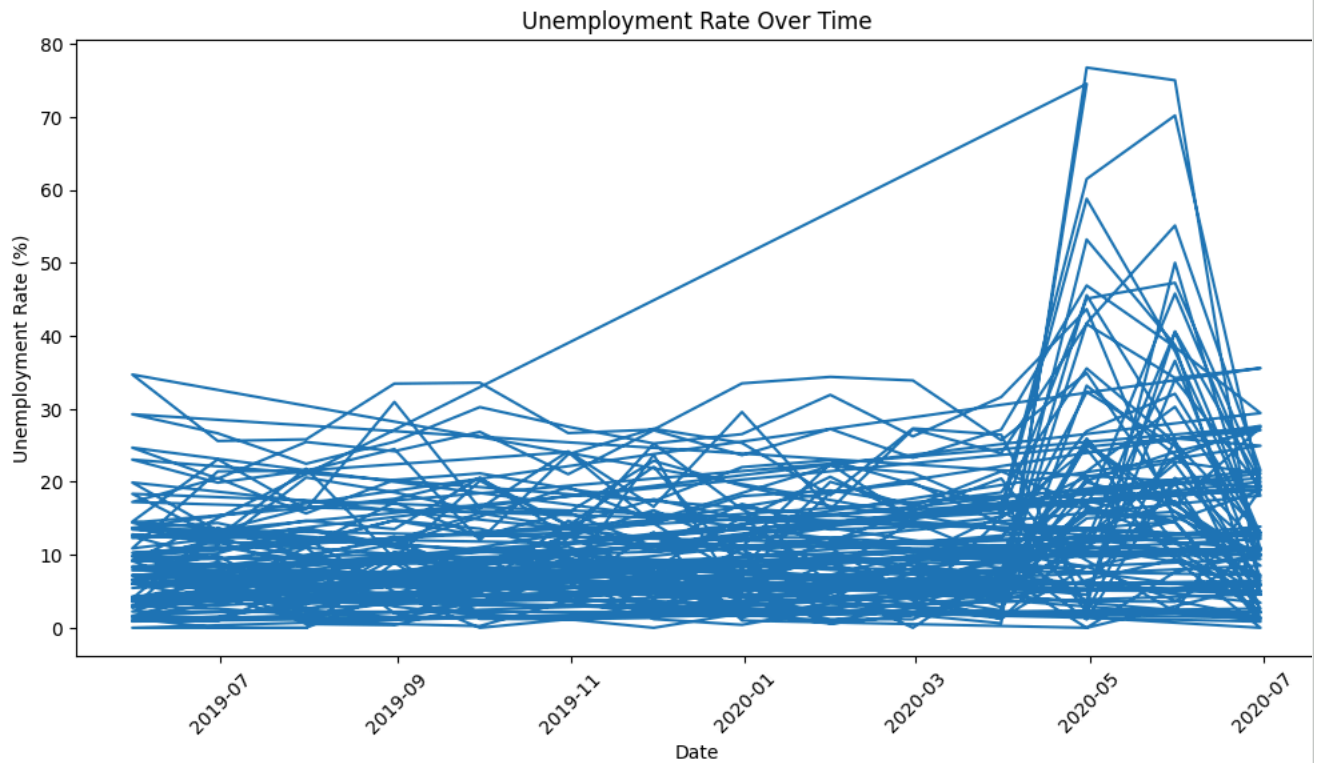
plt.plot(data['Date'], data['Unemployment_Rate'])

plt.title('Unemployment Rate Over Time')
plt.xlabel('Date')
plt.ylabel('Unemployment Rate (%)')

```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



COVID-19 Impact Analysis

```
covid_period = data[data['Date'] >= '2020-03-01']
```

```
before_covid = data[data['Date'] < '2020-03-01']['Unemployment_Rate'].mean()
```

```
during_covid = covid_period['Unemployment_Rate'].mean()
```

```
print("Before COVID:", before_covid)
```

```
print("During COVID:", during_covid)
```

```
Before COVID: 9.509533582089553
```

```
During COVID: 17.774362745098042
```

```
plt.figure(figsize=(12,6))
```

```
plt.plot(data['Date'], data['Unemployment_Rate'], label='Unemployment Rate')
```

```
plt.axvline(pd.to_datetime('2020-03-01'),  
            color='red',  
            linestyle='--',  
            label='COVID-19 Start')
```

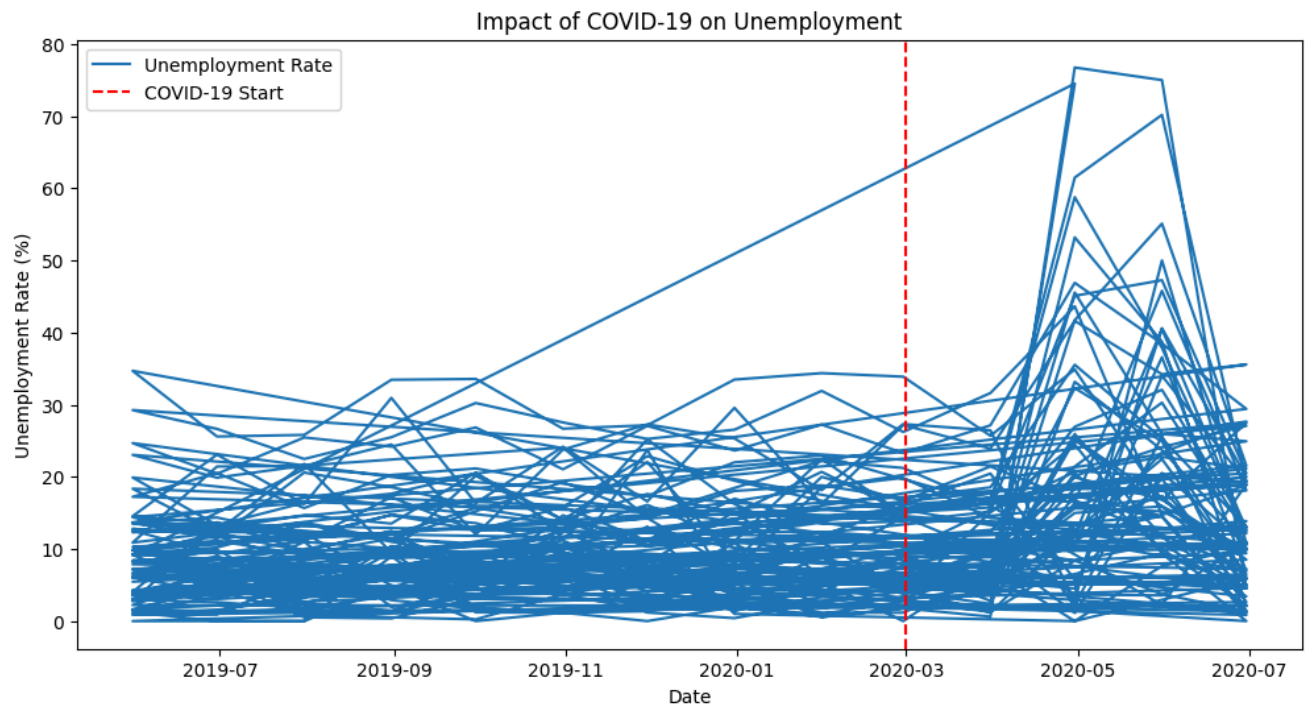
```
plt.legend()
```

```
plt.title('Impact of COVID-19 on Unemployment')
```

```
plt.xlabel('Date')
```

```
plt.ylabel('Unemployment Rate (%)')
```

```
plt.show()
```



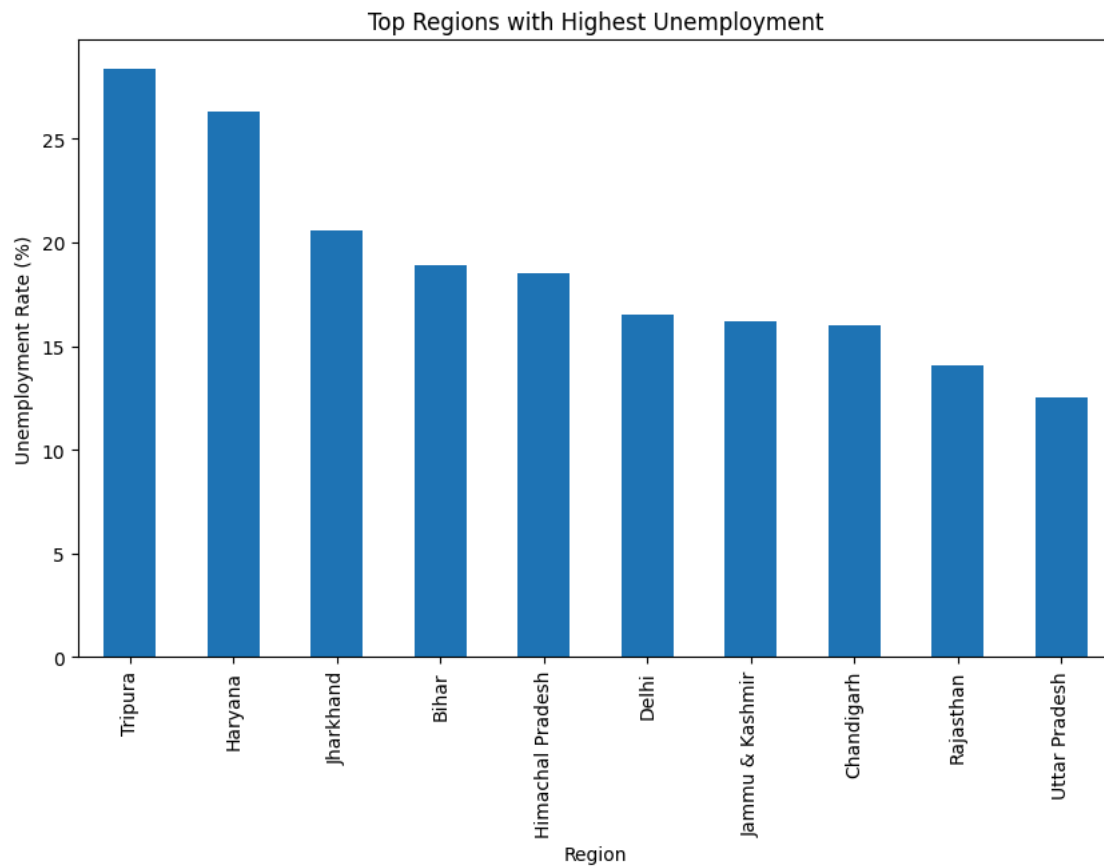
Regional Analysis

```
top_regions = data.groupby('Region')['Unemployment_Rate'].mean().sort_values(ascending=False).head(10)

top_regions.plot(kind='bar', figsize=(10,6))

plt.title('Top Regions with Highest Unemployment')
plt.ylabel('Unemployment Rate (%)')

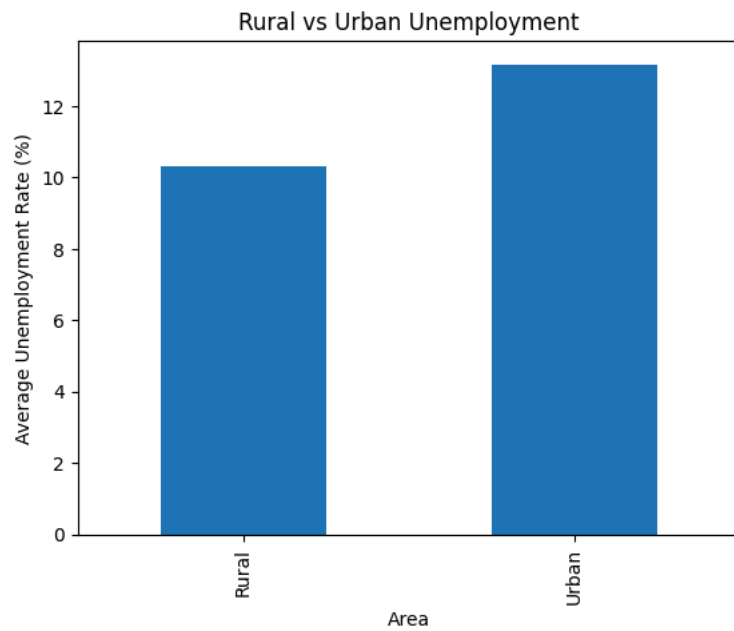
plt.show()
```



```
area_analysis = data.groupby('Area')['Unemployment_Rate'].mean()  
print(area_analysis)
```

```
Area  
Rural    10.324791  
Urban    13.166614  
Name: Unemployment_Rate, dtype: float64
```

```
area_analysis.plot(kind='bar')  
  
plt.title('Rural vs Urban Unemployment')  
plt.ylabel('Average Unemployment Rate (%)')  
  
plt.show()
```



Seasonal Trend Analysis

```
data['Month'] = data['Date'].dt.month
```

```
monthly_trend = data.groupby('Month')['Unemployment_Rate'].mean()  
print(monthly_trend)
```

```
Month  
1    9.950755  
2    9.964717  
3   10.700577  
4   23.641569  
5   16.646190  
6   10.553462  
7    9.033889  
8    9.637925  
9    9.051731  
10   9.900909  
11   9.868364  
12   9.497358  
Name: Unemployment_Rate, dtype: float64
```

```
monthly_trend.plot(marker='o', figsize=(10,5))  
  
plt.title('Seasonal Unemployment Trend')  
  
plt.xlabel('Month')  
plt.ylabel('Average Unemployment Rate (%)')  
  
plt.grid()  
  
plt.show()
```

Seasonal Unemployment Trend

